

Module 03: Perception — Predictive Coding, Sensory Attenuation, and Hallucinations

Learning Objectives

1. Describe the **predictive coding** architecture: how hierarchical cortical layers generate top-down predictions and bottom-up prediction errors.
2. Explain how sensory attenuation (reduced perception of self-generated stimuli) demonstrates the action-perception duality.
3. Analyze hallucinations in schizophrenia and other conditions as failures of the precision-weighted prediction error mechanism.

Introduction

Rao and Ballard (1999) demonstrated that hierarchical predictive coding in the visual cortex could explain both classical receptive field properties and context-dependent response modulation. Their model established a neural architecture for Active Inference's claim that perception is inference: higher cortical areas send **predictions** down the hierarchy; lower areas send **prediction errors** up. Perception occurs when these prediction errors are minimized.

Key Concepts

1. The Predictive Coding Architecture

The six-layered neocortex implements a message-passing scheme:

- **Deep pyramidal cells** (layers 5/6) send **top-down predictions** to lower cortical areas
- **Superficial pyramidal cells** (layers 2/3) send **bottom-up prediction errors** to higher areas
- **Granular layer** (layer 4) receives both streams

In V1, this means that the response of a neuron to a visual stimulus depends not only on what is in its receptive field but on whether the stimulus was *predicted*. Expected stimuli produce

attenuated responses (the prediction was accurate); unexpected stimuli produce amplified responses (large prediction error).

2. Sensory Attenuation

When you try to tickle yourself, it doesn't work. This is **sensory attenuation** — the brain predicts the sensory consequences of its own actions and attenuates the corresponding prediction errors. The cerebellum sends an **efferece copy** of the motor command to sensory cortex, enabling the brain to "cancel out" expected sensory feedback.

This is direct evidence for the Active Inference claim that action and perception share the same inferential machinery. It also explains why psychosis can involve a failure of sensory attenuation — if the brain does not properly predict its own actions, self-generated stimuli (inner speech, for example) may be experienced as externally generated (auditory hallucinations).

3. Hallucinations as Aberrant Inference

Hallucinations can be understood as the brain's own predictions becoming so strong (high precision) that they override incoming sensory evidence. Powers et al. (2017) showed that individuals prone to hallucinations are also more susceptible to *Pavlovian* hallucination induction — they "hear" a tone that was paired with a visual stimulus, even when the tone is absent. This suggests a general precision imbalance: hallucinatory brains overweight top-down predictions relative to bottom-up sensory evidence.

Clinical Connections

- **Schizophrenia:** Auditory hallucinations may reflect failed sensory attenuation of inner speech, combined with high precision on endogenous predictions.
- **Charles Bonnet syndrome:** Visual hallucinations in people with visual loss occur because the brain compensates for reduced sensory input by amplifying its own visual predictions.

Conclusion

Predictive coding provides the neural implementation of Active Inference's central claim: perception is not passive recording but active construction through precision-weighted prediction error minimization. Module 04 examines the cognitive processes that modulate this machinery — attention, neuromodulation, and precision weighting.